the path difference must be, $\delta = 0, \lambda, 2\lambda \ldots$ In general, the integral multiples of λ.

$$\delta = n\lambda \text{ where, } n = 0, 1, 2, 3 \ldots \quad (7.24)$$

For destructive interference, the phase difference should be, $\phi = \pi, 3\pi, 5\pi \ldots$ Hence, the path difference must be, $\delta = \frac{\lambda}{2}, \frac{3\lambda}{2} \ldots\ldots$ In general, the half integral multiples of λ.

$$\delta = (2n-1)\frac{\lambda}{2} \text{ where, } n = 1, 2, 3 \ldots \quad (7.25)$$

EXAMPLE 7.5

The wavelength of a light is 450 nm. How much phase it will differ for a path of 3 mm?

Solution

Wavelength, $\lambda = 450 \text{ nm} = 450 \times 10^{-9} \text{ m}$

Path difference, $\delta = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$

Relation between phase difference and path difference, $\phi = \frac{2\pi}{\lambda} \times \delta$

Substituting,

$$\phi = \frac{2\pi}{450 \times 10^{-9}} \times 3 \times 10^{-3} = \frac{\pi}{75} \times 10^{6}$$

$$\phi = \frac{\pi}{75} \times 10^{6} \text{ rad} = 4.19 \times 10^{4} \text{ rad.}$$

7.3.2 Coherent sources

Two light sources are said to be coherent if they produce waves which have same phase or constant phase difference, same frequency or wavelength (monochromatic), same waveform and preferably same amplitude. Coherence is a property of waves that enables to obtain stationary interference patterns.

Two independent monochromatic sources can never be coherent, because they may emit waves of same frequency and same amplitude, but not with same phase. This is because, atoms while emitting light, produce change in phase due to thermal vibrations. Hence, these sources are said to be incoherent sources.

To obtain coherent light waves, we have the following three techniques.

(i) Wavefront division

(ii) Intensity (or) Amplitude division

(iii) Source and Images.

(i) Wavefront division: This is the most commonly used method for producing coherent sources. We know a point source produces spherical wavefronts. All the points on the wavefront are at the same phase. If two points are chosen on the wavefront by using a double slit, the two points will act as coherent sources as shown in Figure 7.8.

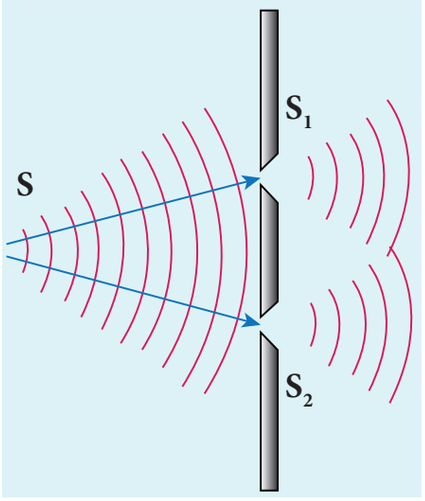


Figure 7.8 Wavefront division

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